LESSON PLAN

MOCKUP SQUADRON ENVIRONMENT (MSE) (24)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	19
Total No. of Lesson Plans	21	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Handling of devices and assemblies and Electrical testing precaution
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Handling of devices and assemblies and Electrical testing precaution.
5. Specific Objectives	24.15.1.1 Trainee will be able to follow Electrostatic sensitive device handling practices in a workshop.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of ESD wrist strap connection and verification using a test meter; Instructor supervision of trainee donning wrist straps and handling ESD sensitive components on a grounded workbench.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary risk of static electricity to electronic components?
(b) Why must a workbench be grounded before handling sensitive devices?
(c) What personal protective equipment is required for ESD sensitive work?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Electrostatic Discharge (ESD) • Sources of static charge in workshop environment • Consequences of ESD damage to assemblies • Standard ESD handling procedures for devices • Proper use of wrist straps and grounding mats	L & Ds	PPT, IB	15 min
(b)	• Electrical testing safety precautions • Verification of zero voltage before touching terminals • Use of insulated tools and personal protective gear • Lockout/Tagout procedures for power sources • Correct sequence for connecting test leads	L & Ds	PPT, PDF	20 min
(c)	• Storage requirements for ESD sensitive assemblies • Identification of ESD safe packaging materials • Procedures for transporting devices within workshop • Inspection checklist for damaged packaging • Reporting protocol for suspected ESD damage	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the correct method for donning an ESD wrist strap.
(b) List three critical precautions to take before performing electrical testing.
(c) Describe the immediate action required if a device shows signs of ESD damage.

12. Home Assignment and Follow Up	Prepare a brief report outlining the specific steps required to set up a compliant Special Handling Area for storage, as covered in the next lesson plan.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Special Handling area and storage.	Refer to the master syllabus reference list	LP 20

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min